

Nonlinear Recurrences, Fixed Points, and Metric-Space Exercises

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§1.1 Fixed-point iteration

An iterative sequence is organized around fixed points. If a function f has a fixed point x_0 , meaning

$$f(x_0) = x_0,$$

then the recurrence

$$a_{n+1} = f(a_n)$$

may converge to x_0 , depending on the behavior of f . The graphical picture is the intersection of $y = f(x)$ and $y = x$.

The elementary idea is this: if $a_n \rightarrow L$ and f is continuous, then

$$L = \lim a_{n+1} = \lim f(a_n) = f(L),$$

so any limit must be a fixed point. But finding a fixed point is not the same as proving convergence. A recurrence can have fixed points and still oscillate or diverge.

§1.2 Möbius recurrences

The page considers recurrences of the form

$$a_{n+1} = \frac{pa_n + q}{ra_n + t}.$$

These are fractional linear or Möbius transformations. Their fixed points solve

$$x = \frac{px + q}{rx + t},$$

so

$$rx^2 + (t - p)x - q = 0.$$

Suppose x_1, x_2 are two distinct fixed points. Then a standard calculation gives

$$\frac{a_{n+1} - x_1}{a_{n+1} - x_2} = \lambda \frac{a_n - x_1}{a_n - x_2}$$

for a constant λ . Thus the new sequence

$$b_n = \frac{a_n - x_1}{a_n - x_2}$$

is geometric.

This is the key idea in the note: convert a nonlinear recurrence into a geometric progression by using the fixed points as coordinates. This is not accidental. Möbius transformations act naturally on the projective line, and the cross-ratio-like expression above is the coordinate that linearizes the action.

§1.3 Example

For

$$a_1 = \frac{1}{2}, \quad a_{n+1} = \frac{5a_n - 1}{a_n + 3},$$

the fixed-point equation is

$$x = \frac{5x - 1}{x + 3},$$

so

$$x^2 - 2x + 1 = 0, \quad x = 1.$$

Here the two fixed points coincide, so the previous method degenerates. One then studies

$$a_{n+1} - 1 = \frac{4a_n - 4}{a_n + 3} = \frac{4(a_n - 1)}{a_n + 3}.$$

Taking reciprocals gives

$$\frac{1}{a_{n+1} - 1} = \frac{a_n + 3}{4(a_n - 1)} = \frac{1}{4} + \frac{1}{a_n - 1}.$$

Thus

$$b_n = \frac{1}{a_n - 1}$$

is arithmetic, not geometric. This is the repeated-root analogue of the fixed-point method.

§1.4 Second-order linear recurrences

The next topic is

$$a_{n+2} = pa_{n+1} + qa_n.$$

Try $a_n = \lambda^n$. The characteristic equation is

$$\lambda^2 - p\lambda - q = 0.$$

If it has two distinct roots λ_1, λ_2 , then

$$a_n = A\lambda_1^n + B\lambda_2^n.$$

If the root is repeated, then the second independent solution is $n\lambda^n$, so

$$a_n = (A + Bn)\lambda^n.$$

This is the recurrence analogue of solving a linear differential equation with constant coefficients.

§1.5 Dedekind cuts and metric exercises

The uploaded pages also continue exercises on Dedekind cuts and metric spaces. For Dedekind cuts X, Y , the sum is

$$X + Y = \{x + y : x \in X, y \in Y\}.$$

The notes check that this is again a cut: it is nonempty, not all of $\mathbb{Q}$, downward closed, and has no largest rational.

For a metric space (X, d) , if $Y \subseteq X$, the subspace metric is

$$d_Y(y_1, y_2) = d(y_1, y_2).$$

The note also checks that Euclidean distance on $\mathbb{R}^n$

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}$$

is a metric. The only nontrivial axiom is the triangle inequality, which ultimately follows from Cauchy–Schwarz.

§1.6 Fixed points of recurrence maps

A recurrence $x_{n+1} = f(x_n)$ is an iteration problem. A possible limit L must satisfy

$$L = f(L).$$

Thus the first step is to solve the fixed-point equation. The second step is to decide whether the iteration actually converges to that fixed point.

A common local test is $|f'(L)| < 1$. Then nearby points are pulled toward L . If $|f'(L)| > 1$, nearby points are pushed away. This explains why fixed points are not automatically limits.

§1.7 Monotone bounded sequences

Many elementary recurrence proofs use the theorem: a monotone bounded real sequence converges. The method is to prove by induction that the sequence stays in an interval, then prove monotonicity. Completeness of $\mathbb{R}$ is the hidden reason the limit exists.

§1.8 After the examples

The important common thread is change of coordinates. A Möbius recurrence becomes geometric after measuring position relative to fixed points. A repeated fixed point becomes arithmetic after taking a reciprocal. A Dedekind cut becomes a real number after using order-theoretic structure. A subset becomes a metric space after inheriting distance. The mathematical move is the same: choose the representation in which the structure becomes visible.

§1.9 Fixed points and stability

For an iteration $x_{n+1} = F(x_n)$, a fixed point satisfies $F(x_*) = x_*$. The behavior near x_* is controlled by $F'(x_*)$ in one dimension: if $|F'(x_*)| < 1$, the fixed point is locally attracting.

§1.10 Möbius recurrences and matrices

A recurrence of the form

$$x_{n+1} = \frac{ax_n + b}{cx_n + d}$$

is a fractional-linear transformation. It corresponds to the matrix

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

acting on the projective line. Iterating the recurrence corresponds to taking powers of this matrix.

§1.11 Second-order recurrences

A linear recurrence

$$a_{n+2} = pa_{n+1} + qa_n$$

is governed by the characteristic equation

$$\lambda^2 = p\lambda + q.$$

Distinct roots give combinations of geometric sequences. This is diagonalization of the shift dynamics.

§1.12 Recurrences as one-dimensional dynamics

Nonlinear and linear recurrences are dynamical systems. Fixed points organize stability, Möbius recurrences are projective matrix dynamics, and second-order linear recurrences are solved by eigenvalue methods.